Lesson 06 material: Designing a Study, Analyzing Data and Reporting Results

2.6: Designing a Research Study

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Learning Objectives

1. Define the concept of a variable, distinguish quantitative from categorical variables, and give examples of variables that might be of interest to psychologists.
2. Explain the difference between a population and a sample.
3. Distinguish between experimental and non-experimental research.
4. Distinguish between lab studies, field studies, and field experiments.

Identifying and Defining the Variables and Population

Variables and Operational Definitions

Part of generating a hypothesis involves identifying the variables that you want to study and operationally defining those variables so that they can be measured. Research questions in psychology are about variables. A **variable** is a quantity or quality that varies across people or situations. For example, the height of the students enrolled in a university course is a variable because it varies from student to student. The chosen major of the students is also a variable as long as not everyone in the class has declared the same major. Almost everything in our world varies and as such thinking of examples of constants (things that don't vary) is far more difficult. A rare example of a constant is the speed of light. Variables can be either quantitative or categorical. A **quantitative variable** is a quantity, such as height, that is typically measured by assigning a number to each individual. Other examples of quantitative variables include people's level of talkativeness, how depressed they are, and the number of siblings they have. A **categorical variable** is a quality, such as chosen major, and is typically measured by assigning a category label to each individual (e.g., Psychology, English, Nursing, etc.). Other examples include people's nationality, their occupation, and whether they are receiving psychotherapy.

After the researcher generates their hypothesis and selects the variables they want to manipulate and measure, the researcher needs to find ways to actually measure the variables of interest. This requires an **operational definition**—a definition of the variable in terms of precisely how it is to be measured. Most variables that researchers are interested in studying cannot be directly observed or measured and this poses a problem because empiricism (observation) is at the heart of the scientific method. Operationally defining a variable involves taking a abstract construct like depression that cannot be directly observed and transforming it into something that can be directly observed and measured. Most variables can be operationally defined in many different ways. For example, depression can be operationally defined as people's scores on a paper-and-pencil depression scale such as the Beck Depression Inventory, the number of depressive symptoms they are experiencing, or whether they have been diagnosed with major depressive disorder. Researchers are wise to choose an operational definition that has been used extensively in the research literature.

Sampling and Measurement

In addition to identifying which variables to manipulate and measure, and operationally defining those variables, researchers need to identify the population of interest. Researchers in psychology are usually interested in drawing conclusions about some very large group of people. This is called the **population**. It could be all American teenagers, children with autism, professional athletes, or even just human beings—depending on the interests and goals of the researcher. But they usually study only a small subset or **sample** of the population. For example, a researcher might measure the talkativeness of a few hundred university students with the intention of drawing conclusions about the talkativeness of men and women in general. It is important, therefore, for researchers to use a representative sample—one that is similar to the population in important respects.

One method of obtaining a sample is **simple random sampling**, in which every member of the population has an equal chance of being selected for the sample. For example, a pollster could start with a list of all the registered voters in a city (the population), randomly select 100 of them from the list (the sample), and ask those 100 whom they intend to vote for. Unfortunately, random sampling is difficult or impossible in most psychological research because the populations are less clearly defined than the registered voters in a city. How could a researcher give all American teenagers or all children with autism an equal chance of being selected for a sample? The most common alternative to random sampling is **convenience sampling**, in which the sample consists of individuals who happen to be nearby and willing to participate (such as introductory psychology students). Of course, the obvious problem with convenience sampling is that the sample might not be representative of the population and therefore it may be less appropriate to generalize the results from the sample to that population.

Experimental vs. Non-Experimental Research

The next step a researcher must take is to decide which type of approach they will use to collect the data. As you will learn in your research methods course there are many different approaches to research that can be divided in many different ways. One of the most fundamental distinctions is between experimental and non-experimental research.

Experimental Research

Researchers who want to test hypotheses about causal relationships between variables (i.e., their goal is to explain) need to use an experimental method. This is because the experimental method is the only method that allows us to determine causal relationships. Using the experimental approach,

researchers first manipulate one or more variables while attempting to control extraneous variables, and then they measure how the manipulated variables affect participants' responses.

The terms independent variable and dependent variable are used in the context of experimental research. The **independent variable** is the variable the experimenter manipulates (it is the presumed cause) and the **dependent variable** is the variable the experimenter measures (it is the presumed effect).

Extraneous variables are any variable other than the dependent variable. **Confounds** are a specific type of extraneous variable that systematically varies along with the variables under investigation and therefore provides an alternative explanation for the results. When researchers design an experiment they need to ensure that they control for confounds; they need to ensure that extraneous variables don't become confounding variables because in order to make a causal conclusion they need to make sure alternative explanations for the results have been ruled out.

As an example, if we manipulate the lighting in the room and examine the effects of that manipulation on workers' productivity, then the lighting conditions (bright lights vs. dim lights) would be considered the independent variable and the workers' productivity would be considered the dependent variable. If the bright lights are noisy then that noise would be a confound since the noise would be present whenever the lights are bright and the noise would be absent when the lights are dim. If noise is varying systematically with light then we wouldn't know if a difference in worker productivity across the two lighting conditions is due to noise or light. So confounds are bad, they disrupt our ability to make causal conclusions about the nature of the relationship between variables. However, if there is noise in the room both when the lights are on and when the lights are off then noise is merely an extraneous variable (it is a variable other than the independent or dependent variable) and we don't worry much about extraneous variables. This is because unless a variable varies systematically with the manipulated independent variable it cannot be a competing explanation for the results.

Non-Experimental Research

Researchers who are simply interested in describing characteristics of people, describing relationships between variables, and using those relationships to make predictions can use non-experimental research. Using the non-experimental approach, the researcher simply measures variables as they naturally occur, but they do not manipulate them. For instance, if I just measured the number of traffic fatalities in America last year that involved the use of a cell phone but I did not actually manipulate cell phone use then this would be categorized as non-experimental research. Alternatively, if I stood at a busy intersection and recorded drivers' genders and whether or not they were using a cell phone when they passed through the intersection to see whether men or women are more likely to use a cell phone when driving, then this would be non-experimental research. It is important to point out that non-experimental does not mean nonscientific. Non-experimental research is scientific in nature. It can be used to fulfill two of the three goals of science (to describe and to predict). However, unlike with experimental research, we cannot make causal conclusions using this method; we cannot say that one variable causes another variable using this method.

Laboratory vs. Field Research

The next major distinction between research methods is between laboratory and field studies. A **laboratory study** is a study that is conducted in the laboratory environment. In contrast, a **field study** is a study that is conducted in the real-world, in a natural environment.

Laboratory experiments typically have high **internal validity**. Internal validity refers to the degree to which we can confidently infer a causal relationship between variables. When we conduct an experimental study in a laboratory environment we have very high internal validity because we manipulate one variable while controlling all other outside extraneous variables. When we manipulate an independent variable and observe an effect on a dependent variable and we control for everything else so that the only difference between our experimental groups or conditions is the one manipulated variable then we can be quite confident that it is the independent variable that is causing the change in the dependent variable. In contrast, because field studies are conducted in the real-world, the experimenter typically has less control over the environment and potential extraneous variables, and this decreases internal validity, making it less appropriate to arrive at causal conclusions.

But there is typically a trade-off between internal and external validity. **External validity** simply refers to the degree to which we can generalize the findings to other circumstances or settings, like the real-world environment. When internal validity is high, external validity tends to be low; and when internal validity is low, external validity tends to be high. So laboratory studies are typically low in external validity, while field studies are typically high in external validity. Since field studies are conducted in the real-world environment it is far more appropriate to generalize the findings to that real-world environment than when the research is conducted in the more artificial sterile laboratory.

Finally, there are field studies which are non-experimental in nature because nothing is manipulated. But there are also **field experiments** where an independent variable is manipulated in a natural setting and extraneous variables are controlled. Depending on their overall quality and the level of control of extraneous variables, such field experiments can have high external and high internal validity.

2.7: Analyzing the Data

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Learning Objectives

1. Distinguish between descriptive and inferential statistics
2. Identify the different kinds of descriptive statistics researchers use to summarize their data
3. Describe the purpose of inferential statistics.
4. Distinguish between Type I and Type II errors.

Once the study is complete and the observations have been made and recorded the researchers need to analyze the data and draw their conclusions. Typically, data are analyzed using both descriptive and inferential statistics. Descriptive statistics are used to summarize the data and inferential statistics are used to generalize the results from the sample to the population. In turn, inferential statistics are used to make conclusions about whether or not a theory has been supported, refuted, or requires modification.

Descriptive Statistics

Descriptive statistics are used to organize or summarize a set of data. Examples include percentages, measures of central tendency (mean, median, mode), measures of dispersion (range, standard deviation, variance), and correlation coefficients.

Measures of central tendency are used to describe the typical, average and center of a distribution of scores. The **mode** is the most frequently occurring score in a distribution. The **median** is the midpoint of a distribution of scores. The **mean** is the average of a distribution of scores.

Measures of dispersion are also considered descriptive statistics. They are used to describe the degree of spread in a set of scores. So are all of the scores similar and clustered around the mean or is there a lot of variability in the scores? The **range** is a measure of dispersion that measures the distance between the highest and lowest scores in a distribution. The **standard deviation** is a more sophisticated measure of dispersion that measures the average distance of scores from the mean. The **variance** is just the standard deviation squared. So it also measures the distance of scores from the mean but in a different unit of measure.

Typically means and standard deviations are computed for experimental research studies in which an independent variable was manipulated to produce two or more groups and a dependent variable was measured quantitatively. The means from each experimental group or condition are calculated separately and are compared to see if they differ.

For non-experimental research, simple percentages may be computed to describe the percentage of people who engaged in some behavior or held some belief. But more commonly non-experimental research involves computing the correlation between two variables. A **correlation coefficient** describes the strength and direction of the relationship between two variables. The values of a correlation coefficient can range from –1.00 (the strongest possible negative relationship) to +1.00 (the strongest possible positive relationship). A value of 0 means there is no relationship between the two variables. Positive correlation coefficients indicate that as the values of one variable increase, so do the values of the other variable. A good example of a positive correlation is the correlation between height and weight, because as height increases weight also tends to increase. Negative correlation coefficients indicate that as the value of one variable increase, the values of the other variable decrease. An example of a negative correlation is the correlation between stressful life events and happiness; because as stress increases, happiness is likely to decrease.

Inferential Statistics

As you learned in the section of this chapter on sampling, typically researchers sample from a population but ultimately they want to be able to generalize their results from the sample to a broader population. Researchers typically want to infer what the population is like based on the sample they studied. **Inferential statistics** are used for that purpose. Inferential statistics allow researchers to draw conclusions about a population based on data from a sample. Inferential statistics are crucial because the effects (i.e., the differences in the means or the correlation coefficient) that researchers find in a study may be due simply to random chance variability or they may be due to a real effect (i.e., they may reflect a real relationship between variables or a real effect of an independent variable on a dependent variable).

Researchers use inferential statistics to determine whether their effects are statistically significant. A **statistically significant** effect is one that is unlikely due to random chance and therefore likely represents a real effect in the population. More specifically results that have less than a 5% chance of being due to random error are typically considered statistically significant. When an effect is statistically significant it is appropriate to generalize the results from the sample to the population. In contrast, if inferential statistics reveal that there is more than a 5% chance that an effect could be due to chance error alone then the researcher must conclude that their result is not statistically significant.

It is important to keep in mind that statistics are probabilistic in nature. They allow researchers to determine whether the chances are low that their results are due to random error, but they don't provide any absolute certainty. Hopefully, when we conclude that an effect is statistically significant it is a real effect that we would find if we tested the entire population. And hopefully when we conclude that an effect is not statistically significant there really is no effect and if we tested the entire population we would find no effect. And that 5% threshold is set at 5% to ensure that there is a high probability that we make a correct decision and that our determination of statistical significance is an accurate reflection of reality.

But mistakes can always be made. Specifically, two kinds of mistakes can be made. First, researchers can make a **Type I error**, which is a false positive. It is when a researcher concludes that their results are statistically significant (so they say there is an effect in the population) when in reality there is no real effect in the population and the results are just due to chance (they are a fluke). When the threshold is set to 5%, which is the convention, then the researcher has a 5% chance or less of making a Type I error. You might wonder why researchers don't set it even lower to reduce the chances of making a

Type I error. The reason is when the chances of making a Type I error are reduced, the chances of making a Type II error are increased. A **Type II error** is a missed opportunity. It is when a researcher concludes that their results are not statistically significant when in reality there is a real effect in the population and they just missed detecting it. Once again, these Type II errors are more likely to occur when the threshold is set too low (e.g., set at 1% instead of 5%) and/or when the sample was too small.

2.8: Drawing Conclusions and Reporting the Results

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Learning Objectives

1. Identify the conclusions researchers can make based on the outcome of their studies.
2. Describe why scientists avoid the term "scientific proof."
3. Explain the different ways that scientists share their findings.

Drawing Conclusions

Since statistics are probabilistic in nature and findings can reflect type I or type II errors, we cannot use the results of a single study to conclude with certainty that a theory is true. Rather theories are supported, refuted, or modified based on the results of research.

If the results are statistically significant and consistent with the hypothesis and the theory that was used to generate the hypothesis, then researchers can conclude that the theory is supported. Not only did the theory make an accurate prediction, but there is now a new phenomenon that the theory accounts for. If a hypothesis is disconfirmed in a systematic empirical study, then the theory has been weakened. It made an inaccurate prediction, and there is now a new phenomenon that it does not account for.

Although this seems straightforward, there are some complications. First, confirming a hypothesis can strengthen a theory but it can never prove a theory. In fact, scientists tend to avoid the word "prove" when talking and writing about theories. One reason for this avoidance is that the result may reflect a type I error. Another reason for this avoidance is that there may be other plausible theories that imply the same hypothesis, which means that confirming the hypothesis strengthens all those theories equally. A third reason is that it is always possible that another test of the hypothesis or a test of a new hypothesis derived from the theory will be disconfirmed. This difficulty is a version of the famous philosophical "problem of induction." One cannot definitively prove a general principle (e.g., "All swans are white.") just by observing confirming cases (e.g., white swans)—no matter how many. It is always possible that a disconfirming case (e.g., a black swan) will eventually come along. For these reasons, scientists tend to think of theories—even highly successful ones—as subject to revision based on new and unexpected observations.

A second complication has to do with what it means when a hypothesis is disconfirmed. According to the strictest version of the hypothetico-deductive method, disconfirming a hypothesis disproves the theory it was derived from. In formal logic, the premises "if *A* then *B* " and "not *B* " necessarily lead to the conclusion "not *A* ." If *A* is the theory and *B* is the hypothesis ("if *A* then *B* "), then disconfirming the hypothesis ("not *B* ") must mean that the theory is incorrect ("not *A* "). In practice, however, scientists do not give up on their theories so easily. One reason is that one disconfirmed hypothesis could be a missed opportunity (the result of a type II error) or it could be the result of a faulty research design. Perhaps the researcher did not successfully manipulate the independent variable or measure the dependent variable.

A disconfirmed hypothesis could also mean that some unstated but relatively minor assumption of the theory was not met. For example, if Zajonc had failed to find social facilitation in cockroaches, he could have concluded that drive theory is still correct but it applies only to animals with sufficiently complex nervous systems. That is, the evidence from a study can be used to modify a theory. This practice does not mean that researchers are free to ignore disconfirmations of their theories. If they cannot improve their research designs or modify their theories to account for repeated disconfirmations, then they eventually must abandon their theories and replace them with ones that are more successful.

The bottom line here is that because statistics are probabilistic in nature and because all research studies have flaws there is no such thing as scientific proof, there is only scientific evidence.

Reporting the Results

The final step in the research process involves reporting the results. As described in the section on Reviewing the Research Literature in this chapter, results are typically reported in peer-reviewed journal articles and at conferences.

The most prestigious way to report one's findings is by writing a manuscript and having it published in a peer-reviewed scientific journal. Manuscripts published in psychology journals typically must adhere to the writing style of the American Psychological Association (APA style). You will likely be learning the major elements of this writing style in this course.

Another way to report findings is by writing a book chapter that is published in an edited book. Preferably the editor of the book puts the chapter through peer review but this is not always the case and some scientists are invited by editors to write book chapters.

A fun way to disseminate findings is to give a presentation at a conference. This can either be done as an oral presentation or a poster presentation. Oral presentations involve getting up in front of an audience of fellow scientists and giving a talk that might last anywhere from 10 minutes to 1 hour (depending on the conference) and then fielding questions from the audience. Alternatively, poster presentations involve summarizing the study on a large poster that provides a brief overview of the purpose, methods, results, and discussion. The presenter stands by their poster for an hour or two and discusses it with people who pass by. Presenting one's work at a conference is a great way to get feedback from one's peers before attempting to undergo the more rigorous peer-review process involved in publishing a journal article.

2.9: Overview of the Scientific Method (Summary)

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Key Takeaways

- Research in psychology can be described by a simple cyclical model. A research question based on the research literature leads to an empirical study, the results of which are published and become part of the research literature.
- The research literature in psychology is all the published research in psychology, consisting primarily of articles in professional journals and scholarly books.
- Early in the research process, it is important to conduct a review of the research literature on your topic to refine your research question, identify appropriate research methods, place your question in the context of other research, and prepare to write an effective research report.
- There are several strategies for finding previous research on your topic. Among the best is using PsycINFO, a computer database that catalogs millions of articles, books, and book chapters in psychology and related fields.
- Research questions expressed in terms of variables and relationships between variables can be suggested by other researchers or generated by asking a series of more general questions about the behavior or psychological characteristic of interest.
- It is important to evaluate how interesting a research question is before designing a study and collecting data to answer it. Factors that affect interestingness are the extent to which the answer is in doubt, whether it fills a gap in the research literature, and whether it has important practical implications.
- It is also important to evaluate how feasible a research question will be to answer. Factors that affect feasibility include time, money, technical knowledge and skill, and access to special equipment and research participants.
- A theory is broad in nature and explains larger bodies of data. A hypothesis is more specific and makes a prediction about the outcome of a particular study.
- Working with theories is not “icing on the cake.” It is a basic ingredient of psychological research.
- Like other scientists, psychologists use the hypothetico-deductive method. They construct theories to explain or interpret phenomena (or work with existing theories), derive hypotheses from their theories, test the hypotheses, and then reevaluate the theories in light of the new results.
- Variables vary across people or situations and may be quantitative (e.g., age) or categorical (e.g., course subject).
- A sample is a small subset of a larger population that is selected to participate in the research study. There are many different ways of sampling participants including convenience sampling and simple random sampling.
- Experimental research involves manipulating an independent variable to observe the effects on a measured dependent variable while non-experimental research involves measuring variables as they naturally occur (i.e., without manipulating anything).
- Research can be conducted in the field or the lab. Laboratory experiments tend to have high internal validity (allowing us to make strong causal conclusions), while field studies often have more external validity (allowing us to generalize to the real world).
- The mean, median, and mode are measures of central tendency used to describe the typical, average, or center scores in a distribution. The range, standard deviation, and variance are measures of how dispersed or spread apart the scores are. Measures of central tendency and dispersion are important descriptive statistics.
- Inferential statistics allow researchers to determine whether their findings are statistically significant, that is, whether they are unlikely to be due to chance alone and therefore are likely to represent a real effect in the population.
- Since statistics are probabilistic in nature we never know if our conclusions are correct. We can make type I errors (concluding an effect is real when it is not) or type II errors (concluding there is no effect when there actually is a real effect in the population).
- Theories can be supported by not proved. Similarly, disconfirming a hypothesis does not necessarily mean that theory has been disproved.
- The final step of the research process involves reporting results at scientific conferences, in journal articles, and/or in books.

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Exercises

- Practice: Find a description of an empirical study in a professional journal or in one of the scientific psychology blogs. Then write a brief description of the research in terms of the cyclical model presented here. One or two sentences for each part of the cycle should suffice.
- Watch the following TED Ed video, in which David H. Schwartz provides an introduction to two types of empirical studies along with some methods that scientists use to increase the reliability of their results:

A YouTube element has been excluded from this version of the text. You can view it online here: https://kpu.pressbooks.pub/psychmethods4e/?p=271

  - Reading in print? Go to https://youtu.be/GUpd2HJHUt8 or scan this QR code with your phone:
- Practice: Use the techniques discussed in this section to find 10 journal articles and book chapters on one of the following research ideas: memory for smells, aggressive driving, the causes of narcissistic personality disorder, the functions of the intraparietal sulcus, or prejudice against the physically handicapped.
- Watch the following video clip produced by UBCiSchool about how to read an academic paper (without losing your mind):

A YouTube element has been excluded from this version of the text. You can view it online here: https://kpu.pressbooks.pub/psychmethods4e/?p=271

  - Reading in print? Go to https://youtu.be/SKxm2HF_-ko or scan this QR code with your phone:
- Practice: Generate three research ideas based on each of the following: informal observations, practical problems, and topics discussed in recent issues of professional journals.
- Practice: Generate an empirical research question about each of the following behaviors or psychological characteristics: long-distance running, getting tattooed, social anxiety, bullying, and memory for early childhood events.
- Practice: Evaluate each of the research questions you generated in Exercise 2 in terms of its interestingness based on the criteria discussed in this section.
- Practice: Find an issue of a journal that publishes short empirical research reports (e.g., *Psychological Science*, *Psychonomic Bulletin and Review*, *Personality and Social Psychology Bulletin*). Pick three studies, and rate each one in terms of how feasible it would be for you to replicate it with the resources available to you right now. Use the following rating scale: (1) You could replicate it essentially as reported. (2) You could replicate it with some simplifications. (3) You could not replicate it. Explain each rating.
- Practice: Find a recent empirical research report in a professional journal. Read the introduction and highlight in different colors descriptions of theories and hypotheses.
- Practice: Using the research article you found in a professional journal identify whether the study was experimental or non-experimental. If it was experimental identify the independent and dependent variables.
- Practice: Using the research article you found in a professional journal identify which descriptive statistics were reported.
- Practice: Describe why theories can be supported but not proved.

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